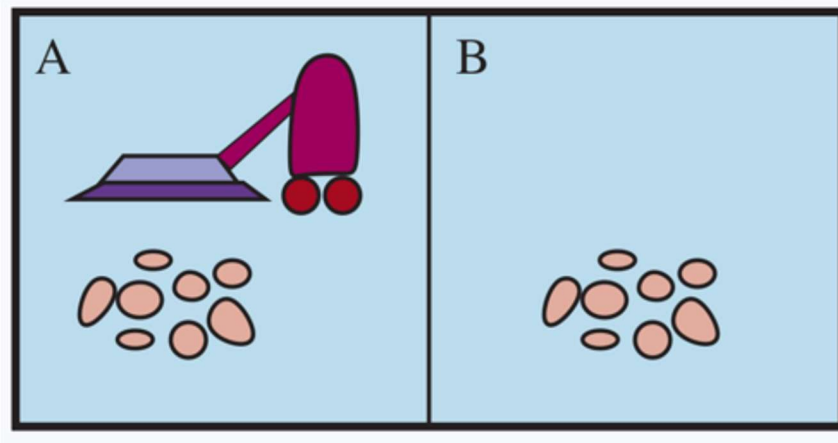


Assignment #1 – Write a Simple Reflex Agent

- Implement a simple reflex agent for the vacuum environment we have talked about. I've provided a description below. *Hint: the pseudocode for this agent is on the slides*
- Run the environment with this agent for all possible initial dirt configurations and agent locations.
- Using a performance measure of one point per clean square at each time step, record the performance score for each configuration.

Task Environment Details:

- Performance Measure: Cleanliness (one point per clean square are each time step)
- Environment: World is divided into two squares A & B.
- Sensors: The agent can assess if the squares are dirty or clean.
- Actuators: The agent can clean the square or move onto the next square.



Programming language: You are welcome to use any programming language you like. This might be a good opportunity for you all to practice with python with a simple agent before we move onto more complex tasks.

Rubric (40 points):

- Implementing the agent (10 points)
 - actions of cleaning/moving
 - sensors assessing if square is dirty or clean
- Creating environment (5 points)
 - Clean or dirty squares
- Calculating performance measure (10 points)
 - One point per clean square at each time step
- Running the environment (10 points)
 - Runs with all possible initial dirt configurations and agent starting points
- Documentation (5 points)
 - Overall comments describing purpose of program with name and date
 - In-line comments